Course Project Proposal: Pairs Trading Strategy Framework with Risk Controls

Nijesh Raghava **Vivek Hruday** **Kushagra Trivedi** **Tanishq Agarwal**

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1 Introduction & Background

Pairs trading is a classic statistical arbitrage strategy. The core premise relies on identifying two historically correlated assets whose prices have temporarily diverged. By taking a long position in the underperforming asset and a short position in the outperforming asset, the strategy bets that the spread between the two will mean-revert to its historical average.

While the underlying statistical concept (cointegration) is well-understood, the practical challenge—and the focus of this project—lies in building a robust framework that can survive real-world frictions. Additionally, this project will explore whether more sophisticated relationship modeling techniques (e.g., causal inference via Granger causality) offer meaningful advantages over traditional correlation-based approaches in pair selection and signal generation.

2 Project Objectives & Scope

This project will contribute a self-contained, interoperable strategy module to the broader Quant Finance Toolkit Ecosystem. Our primary objective is to evaluate whether a statistically sound pairs strategy can survive transaction costs and downside risk scenarios, while also comparing different methodologies for identifying tradable pairs.

In-Scope Deliverables:

- **Pair Identification Engine:** Tools to identify candidate pairs using:
  - Correlation-based screening (baseline approach)
  - Cointegration testing (Engle-Granger, ADF)
  - Causal relationship analysis (Granger causality tests)
- **Signal & Sizing Logic:** A z-score based mean-reversion signal generator coupled with a volatility-targeting position sizer.
- **Risk Management Layer:** Implementation of critical risk controls, including maximum leverage limits, Value-at-Risk (VaR) guardrails, and dynamic stop-losses.
- **Backtesting & Evaluation:** A backtest runner that accounts for transaction costs (bps per trade) and outputs performance metrics (Sharpe ratio, max drawdown, turnover).
- **Methodological Comparison Study:** A structured comparison between correlation-based and causal (Granger-based) pair selection approaches, evaluating differences in:
  - Stability of identified pairs
  - Signal quality and mean-reversion behavior
  - Risk-adjusted returns after costs

3 System Architecture & Tech Stack

To ensure seamless integration with the shared repository framework, our module will be developed using a standardized, object-oriented approach in Python.

- **Language & Core Libraries:** Python, NumPy, Pandas (for vectorized data manipulation and time-series alignment).
- **Statistical Modeling:** Statsmodels and scikit-learn for rolling regressions, cointegration testing, and Granger causality analysis.
- **Data Ingestion:** Yahoo Finance (`yfinance`) and Stooq for historical OHLCV data.
- **Evaluation & Visualization:** Matplotlib and Seaborn for plotting spread distributions, z-scores, drawdown curves, and comparative performance analysis.

4 Timeline & Team Responsibilities

The following 4-week sprint is designed to parallelize development across our 4-member team.

Week 1: Data Pipeline & Statistical Foundations

- **Member 1 & 2:** Build the data downloader, caching mechanism, and the Pair Identification engine (correlation screening, spread computation, ADF/Engle-Granger tests).
- **Member 3 & 4:** Implement Granger causality testing module and define shared `Instrument` and `MarketDataSnapshot` interfaces.

Week 2: Signal Generation & Position Sizing

- **Member 1 & 3:** Develop the z-score threshold logic (entry/exit signals).
- **Member 2 & 4:** Implement the position sizing algorithms, specifically scaling weights inversely to the rolling volatility of the spread.

Week 3: Risk Controls & Backtest Mechanics

- **Member 1 & 4:** Build the transaction cost simulator and the P&L accounting logic.
- **Member 2 & 3:** Program the Risk Management Layer (stop-loss, time stops, and VaR exposure limits).

Week 4: Integration, Testing, & Documentation

- **All Members:** Write invariant tests (e.g., ensuring weights never exceed max leverage, short/long exposures balance out).
- **All Members:** Perform comparative experiments between correlation-based and causal approaches, including out-of-sample validation.
- **All Members:** Finalize the integrated demo notebook and draft the final evaluation report documenting insights from the comparison study.